

Beacon Digest -- 2026-08-31

Linux

Kernel prepatch 7.3-rc1

Source: *LWN.net*

Linus has released 7.3-rc1 and closed the merge window for this release. " Nothing really stands out - except for the fact that it's big. It's not the biggest rc1 we've ever had, but it's certainly up there, at least in number of commits. " He pulled 15,267 commits during this merge window, making i...

DevOps

Tide launched Raziel for AI security. It assumes hackers are inside.

Source: *The New Stack*

Michael Loewy, co-founder of the Tide Foundation, says the security industry was already losing ground to hackers before AI. Everything The post Tide launched Raziel for AI security. It assumes hackers are inside. appeared first on The New Stack .

Security

China-Linked Fire Ant Hijacks Cisco Routers to Steal Credentials and Blind Security Logs

Source: *The Hacker News*

A China-nexus cyber espionage actor tracked as Fire Ant has expanded a long-running campaign beyond VMware hypervisors to compromise Cisco IOS XR routers, Terminal Access Controller Access-Control System (TACACS) servers, and Linux management hosts used to route, authenticate, and manage high-value...

DoJ Corrects China Hacking Claim, Says U.S. Agencies Were Targets, Not Victims

Source: *The Hacker News*

The U.S. Department of Justice (DoJ) on Friday corrected a previously issued press statement that several of its agencies were victims of attacks carried out by

Chinese threat actors, instead now pointing out that they were among those targeted. Last week, the DoJ said the National Aeronautics and S...

TerminalFix Uses Fake Cloudflare CAPTCHAs to Deploy Reverse-Tunnel Backdoor

Source: The Hacker News

Microsoft has disclosed details of a new ClickFix variant, dubbed TerminalFix, that aims to trick users into running a malicious command in Windows Terminal or PowerShell. "While traditional ClickFix campaigns direct victims to the Windows Run dialog, TerminalFix campaigns apply the same technique b...

Five Critical WordPress Plugin and Theme Flaws Enable Site Takeover or RCE

Source: The Hacker News

Multiple critical security flaws have been disclosed in WordPress plugins and themes, including WPMU DEV Dashboard, Avada, TranslatePress, Pods, and GiveWP, that could lead to authentication bypass, account takeover, and arbitrary code execution. The vulnerabilities, according to Wordfence and Patch...

Berlin Refuses to Pay Hackers Who Stole Data From the City's State Network

Source: The Hacker News

Berlin's state government has confirmed that it is the target of an extortion attempt following the August compromise of the city's state administrative network, and said it will not meet the extortionists' demands. The same statement disclosed that forensic work had found further data outflows in t...

Cosmos EVM Flaw Exploited After Cosmos Labs Knew Every Blockchain Running It Was Vulnerable

Source: The Hacker News

Cosmos Labs has warned that a critical balance-handling flaw in the shared Cosmos EVM module was exploited to drain funds from six blockchains between August 20 and August 25, 2026. The vulnerability, designated GHSA-7g4w-cg88-2cq2, is rated Critical by Cosmos Labs and was published without a CVE id...

Attackers Chain Two PaperCut Flaws to Execute Code Without Authentication

Source: The Hacker News

Malicious actors are exploiting a newly patched security flaw in PaperCut NG and MF to execute arbitrary code on susceptible instances, as the company released a

fresh emergency fix with additional hardening. "This vulnerability gives an unauthenticated attacker remote control over PaperCut's trustee...

Android 17 Adds OS-Wide ECH to Hide Website Visits From Network Providers

Source: The Hacker News

Google on Thursday announced new network security protections in Android 17 to bolster connection privacy, address cellular vulnerabilities, and safeguard the privacy of users' home networks. Topping the list is support for Encrypted Client Hello (ECH), a privacy standard that prevents networks from...

ownCloud Flaw Exploited to Steal Nuclear Records From Philippine Research Body

Source: The Hacker News

The U.S. Cybersecurity and Infrastructure Security Agency (CISA) on Thursday added a critical security flaw impacting ownCloud to its Known Exploited Vulnerabilities (KEV) catalog following reports that a Chinese-speaking threat actor weaponized the vulnerability to target a nuclear research body in...

19 Chrome and Edge Extensions Found With Wallet-Stealing and Crypto-Draining Code

Source: The Hacker News

Cybersecurity researchers have discovered a cluster of 18 Google Chrome and one Microsoft Edge extensions that were published over the last six months and harbored wallet secret stealing and cryptocurrency draining capabilities. The extensions, per Socket security researcher Karlo Zanki, share simil...

Two Unitree G1 EDU Humanoid Robot Flaws Enable Root RCE, One Starts Over Bluetooth

Source: The Hacker News

Security researcher Olivier Laflamme has disclosed two independent root remote code execution (RCE) chains affecting the Unitree G1 EDU, including a Bluetooth Low Energy (BLE) path that can reach root on the robot's Locomotion PC. The flaws are tracked as CVE-2026-76639 and CVE-2026-76640, with the...

Key Reasons Why Identity Fabric Matters in 2026

Source: The Hacker News

An Identity Fabric knits fragmented identity systems into a coherent layer that observes how identities behave across applications, APIs, and infrastructure. As

enterprise access spans more cloud services and automated workloads, identity security depends less on static configuration and more on run...

Three CVSS 10.0 ServiceNow Flaws Could Let Unauthenticated Attackers Execute Code and SQL

Source: The Hacker News

ServiceNow has released patches for four security flaws impacting the ServiceNow AI Platform, three of them rated 10.0 on the CVSS scoring system and exploitable, in certain circumstances, by an unauthenticated attacker. The company said it deployed a security update to hosted instances and provided...

China-Made ZBT Routers Ship With Two Implants Giving Unauthenticated Attackers Root Access

Source: The Hacker News

VulnCheck has disclosed two previously undocumented factory implants in firmware for routers built by Shenzhen Zhibotong Electronics (ZBT), each of which gives an unauthenticated remote attacker the ability to run commands as root on affected devices. The implants, named SPEAKINGSTONE and DARKLANTER...

Critical cPanel Flaw Could Let One Hosting Customer Take Root Control of a Whole Server

Source: The Hacker News

cPanel has released patches for a security flaw affecting domain parking and addon domain functionality in cPanel and WebHost Manager (WHM), which could allow code execution as the root user. The vulnerability, assigned the CVE identifier CVE-2026-65643, impacts all supported versions of cPanel & WH...

PaperCut Zero-Day Exploited in Attacks, Affecting All NG and MF Versions

Source: The Hacker News

PaperCut has alerted customers that bad actors are actively exploiting a vulnerability impacting all versions of its PaperCut NG and PaperCut MF print management software in zero-day attacks. The company has released an emergency patch for v25 and v26 to address the issue. It said it's "aware of con...

APT28-Linked HOOKEDGE Backdoor Targets European Government and Diplomatic Organizations

Source: The Hacker News

Cybersecurity researchers have flagged a fresh set of campaigns targeting government and diplomatic organizations in Romania, Spain, and Türkiye between

late September 2025 and early April 2026. These campaigns, per Recorded Future Insikt Group, have led to the deployment of a previously undocumente...

OpenAI Says Reward Hacking Drove AI Agents to Exploit Zero-Days and Breach Hugging Face

Source: The Hacker News

OpenAI on Wednesday revealed that reward hacking was a key driver behind the artificial intelligence (AI)-powered hack of Hugging Face last month, adding that it found evidence of misaligned behavior as early as late May. The incident, the company said, took place during cybersecurity evaluations of...

Next.js Patches Critical AVIF and Windows Flaws Enabling Unauthenticated RCE

Source: The Hacker News

Credit: Hacktron Vercel has released security patches for two critical-severity vulnerabilities in the Next.js web framework, both of which allow unauthenticated remote code execution, one exploitable via specially crafted AVIF image files and the other through a path traversal flaw affecting server...

ThreatsDay: 296K IoT Botnet, 100+ Water Systems Targeted, SharePoint RCE Chain + 27 New Stories

Source: The Hacker News

A fake login page. A fake security scan. A fake productivity app. Apparently, pretending to be useful is still one of the easier ways into a machine. The rest of the week gets stranger: botnets borrowing AI, command traffic hiding in public infrastructure, malicious tools waiting before showing thei...

Amazon Kiro Prompt Injection Can Exfiltrate Sensitive Data Through Kiro Powers

Source: The Hacker News

Cybersecurity researchers have disclosed details of a vulnerability in Amazon Kiro, an artificial intelligence (AI)-powered, agentic integrated development environment (IDE), that could facilitate data exfiltration via prompt injection and Kiro Powers. The security flaw, which does not have a CVE id...

Learn How to Build Security Operations Ready for AI-Powered Attacks

Source: The Hacker News

Security teams have spent years trying to detect threats faster. AI is changing the harder part: how much time defenders have left to act. Advanced AI models can

now help attackers discover vulnerabilities, generate exploit code, and move through weaknesses faster than traditional security processes...

Alleged TeamPCP Hackers Charged in Australia Over Major Supply Chain Attacks

Source: The Hacker News

The Australian Federal Police (AFP) has charged two Western Australian men with a combined total of 14 offences over their alleged role in TeamPCP, the cybercrime group behind the March 2026 compromise of the open-source security scanners Trivy and Checkmarx KICS and the AI gateway LiteLLM. Louis Mi...

What the Data Says About AI in Security Operations in 2026

Source: The Hacker News

AI is officially mainstream in security operations. According to Prophet Security's State of AI in Security Operations 2026 report (produced from ViB's survey of 250+ cybersecurity pros), 40% of security teams now use AI daily. Another 56% are currently testing it out, and only 4% have no plans to a...

Spark RAT Targets Cambodia, Abuses Vulnerable OPSWAT Driver to Disable Security Tools

Source: The Hacker News

Individuals and organizations in Cambodia have emerged as the target of a new campaign that delivers an open-source remote access trojan (RAT) called Spark RAT. "The samples employ diverse lure themes, suggesting an effort to appeal to a broad range of potential victims. These include government not...

GoCaracal Malware Uses Ethereum Smart Contract to Fetch Replacement C2 Address

Source: The Hacker News

Threat actors linked by Arctic Wolf to Dark Caracal with medium confidence deployed a previously undocumented Go-based malware framework, GoCaracal, during a June 2026 intrusion at an unnamed communications organization in Venezuela. GoCaracal provides operators with remote shell access and payload...

New GPUThor Rowhammer Defeats ECC on NVIDIA RTX A6000 to Gain Host Root Access

Source: The Hacker News

Academic researchers have disclosed a Rowhammer attack impacting NVIDIA workstation GPUs with GDDR6 memory that defeats error correction codes (ECC),

the mitigation NVIDIA recommends against GPU Rowhammer, and enables denial-of-service (DoS) and privilege escalation to a root shell. Dubbed GPUThor,...

CISA Adds Six Exploited Flaws to KEV, Including NetScaler, Linux, and SQL Server Bugs

Source: The Hacker News

The U.S. Cybersecurity and Infrastructure Security Agency (CISA) on Wednesday added six flaws to its Known Exploited Vulnerabilities (KEV) catalog, including a high-severity security vulnerability impacting Citrix NetScaler ADC and NetScaler Gateway, citing evidence of active exploitation. The vulne...

FBI Disrupts China-Linked QTFY Infrastructure Used to Steal Data From U.S. Organizations

Source: The Hacker News

The U.S. Department of Justice (DoJ) on Wednesday announced the disruption of two hacking platforms named QScan and QTRouter operated by Chinese threat actors to target critical infrastructure and other sensitive networks in the country. The activity has been attributed to a Chinese state-sponsored...

Nimbus Manticore Expands Toolset With TWOSTROKE-Like Backdoor and SSH Tunneler

Source: The Hacker News

Cybersecurity researchers have discovered additional infrastructure and previously undocumented malware associated with Nimbus Manticore, an Iranian state-sponsored hacking group affiliated with the Islamic Revolutionary Guard Corps (IRGC). Group-IB, in a new analysis published today, described the...

NovaCookies Campaigns Abuse Genuine Docusign Notifications to Steal Microsoft 365 Sessions

Source: The Hacker News

Cybersecurity researchers have disclosed details of a new adversary-in-the-middle (AitM) phishing toolkit called NovaCookies that's used as a proxy to redirect Microsoft 365 sign-ins, while capturing authenticated sessions in the process. In a report shared with The Hacker News ahead of publication,...

CISA Red Team Compromised Two Critical Infrastructure Orgs, One Detected Nothing

Source: The Hacker News

The U.S. Cybersecurity and Infrastructure Security Agency (CISA) has published the results of two red team assessments it conducted simultaneously against two

critical infrastructure organizations, using what it described as similar tradecraft while recording sharply different defensive outcomes. Bo...

Unpatched Kaltura mwEmbed Flaws Could Let Remote Attackers Read Files and Run Code

Source: The Hacker News

The CERT Coordination Center (CERT/CC) has disclosed two unpatched vulnerabilities in Kaltura's HTML5 video player library that allow a remote, unauthenticated attacker to read arbitrary files from a server and execute code on it. The flaws, tracked as CVE-2026-19913 and CVE-2026-19912, both stem fr...

Imagine the SOC Without a Queue: From Alert Backlog to AI Hypothesis Engine

Source: The Hacker News

The SOC we've always known was built around a model that guarantees most of the alert queue will never receive analyst review. There's never time. In a traditional SOC, the typical progression follows a well-known pattern: an alert arrives; a detection engine assigns a severity score. The issue then...

Claude Opus 4.6 Bypasses Gym Booking Limit, Cancels Other Users' Reservations in Tests

Source: The Hacker News

Aikido Security has published research that recreates the Australian gym-booking incident in a synthetic environment, finding that Claude Opus 4.6, running on the OpenClaw agent harness, exploited a client-side-only booking restriction in 9 of 10 runs. The original incident was first reported by ABC...

OpenAI Bans Russian ChatGPT Accounts Used to Run Influence Operation

Source: The Hacker News

OpenAI on Tuesday said it banned a cluster of Russian ChatGPT accounts that used VPNs to bypass access restrictions and run an influence operation, which relied on its artificial intelligence (AI) tool to generate social media posts and comments that were shared on Substack, Telegram, X, Facebook an...

INTERPOL Operation Jackal IV Arrests 58, Identifies 263 in Global Cyber Fraud Crackdown

Source: The Hacker News

An eight-month INTERPOL operation targeting West African organized crime groups has led to arrests of 58 people and the identification of 263 suspects. "The

operation, which brought together 22 countries from six continents, is a response to the escalating global threat posed by West African crimina...

New SLEEPWALKER Backdoor Waits for One Crafted Packet, Then Runs Its Own Bytecode

Source: The Hacker News

An independent malware researcher has documented a previously unreported Windows backdoor, dubbed SLEEPWALKER, that stays inert in memory until a specifically crafted network packet reaches the machine and then runs commands written in a 23-instruction language of its own design. The sample is an un...

Critical Gitea RCE Actively Exploited as Reported Attack Drops Miner-Like Payload

Source: The Hacker News

The U.S. Cybersecurity and Infrastructure Security Agency (CISA) on Tuesday warned of active exploitation efforts targeting a recently patched critical security flaw impacting Gitea. The vulnerability in question is CVE-2026-60004 (CVSS score: 9.8), a case of remote code execution that allows an att...

Fake Apple Support AI Calls Target Stolen-Device Owners for Passcodes and 2FA Codes

Source: The Hacker News

Cybersecurity researchers have disclosed details of a phishing-as-a-service (PhaaS) platform built to strip Apple's Activation Lock from stolen devices, using rented AI voice agents that call theft victims posing as Apple Support and ask for their device passcode. SOCRadar Threat Research Unit (STRU...

U.S. Sanctions Iran-Linked Hackers Behind Critical Infrastructure Breaches

Source: The Hacker News

The U.S. Department of the Treasury has announced fresh sanctions on Iranian cyber actors as part of what it called an "unprecedented, whole-of-government, economic campaign" against the nation and its enablers. "We are launching an economic onslaught against Iran's financial connections around the...

A Malicious Webpage Could Poison Your Local AI Model Behind NVIDIA NemoClaw

Source: The Hacker News

Oasis Security has disclosed a weakness in NVIDIA NemoClaw that could let an attacker-controlled webpage take unauthenticated control of the local Ollama

instance serving an AI agent and plant hidden instructions inside the model itself. The findings were shared with The Hacker News ahead of publica...

WhatsApp Adds Multiple Passkeys for Phishing-Resistant Sign-Ins Across iOS and Android

Source: The Hacker News

Meta on Tuesday announced a set of WhatsApp account security features, including support for multiple passkeys to a single account to help users with both iOS and Android devices sign into their accounts using the phishing-resistant method. The tech giant said more than 1 billion people use a passke...

Marimo Notebook Flaw Could Run MCP Commands Before Cells Execute in Edit Mode

Source: The Hacker News

Marimo has addressed a high-severity security flaw in its notebook software that allowed an attacker to execute an attacker-supplied Model Context Protocol (MCP) command in a specially crafted notebook, according to VulnCheck's CVE Numbering Authority (CNA) record. The CNA record says the command ca...

Mirage2FA Surge Hits 4,500 US and EU Companies, Abusing Microsoft 365 Login Flows

Source: The Hacker News

Thousands of companies have been affected by the Mirage2FA campaign from 2024 to 2026. The commercial phishing-as-a-service toolkit targets Microsoft 365 accounts by abusing legitimate login flows and bypassing two-factor authentication. According to ANY.RUN research, 48% of targeted email addresses...

24 npm Packages Abuse unpkg Mirrors to Host Fake Cloudflare CAPTCHA Pages

Source: The Hacker News

Cybersecurity researchers have disclosed details of a new campaign that uses a cluster of 24 npm packages as free phishing infrastructure for redirecting to ClickFix-style fake CAPTCHA pages. "While the malware is simply a single HTML page inside the npm package, and while downloading it wouldn't do...

E4del and PINHOLE RATs Turn FTP Banners Into Dead Drops for Malware Commands

Source: The Hacker News

Cybersecurity researchers are calling attention to a new campaign that employs FTP banners as dead drop resolvers (DDRs) to deliver two previously unreported

remote access trojans (RATs) tracked as E4del and PINHOLE. While threat actors are known to abuse legitimate services to point to additional c...

Frontier AI: Vulnerability Management's Systemic Revolution

Source: The Hacker News

Vulnerability management has been a staple of security programs since the dawn of the cybersecurity discipline. The symbiotic relationship between vulnerability and patch management teams has also existed for that time and has gone through waves of contention and thankfulness. While this relationshi...

Attackers Target miniOrange SAML Flaws That Can Grant WordPress Admin Access

Source: The Hacker News

Bad actors are attempting to exploit two severe unauthenticated authentication bypasses in the Xecurify miniOrange SAML 2.0 Single Sign On plugin that make it possible for an attacker to sign in as any WordPress user, including administrators. The vulnerabilities, as disclosed by Patchstack, are lis...

Actively Exploited Oracle WebLogic Flaw Lets Unauthenticated Attackers Access Critical Data

Source: The Hacker News

The U.S. Cybersecurity and Infrastructure Security Agency (CISA) on Monday added a maximum-severity security flaw impacting Oracle HTTP Server and Oracle WebLogic Server to its Known Exploited Vulnerabilities (KEV) catalog, citing evidence of active exploitation. The vulnerability, tracked as CVE-20...

DSA-6479-1 roundcube - security update

Source: Debian Security Advisories

<https://security-tracker.debian.org/tracker/DSA-6479-1>

DSA-6478-1 starlette - security update

Source: Debian Security Advisories

<https://security-tracker.debian.org/tracker/DSA-6478-1>

ISC Stormcast For Monday, August 31st, 2026 <https://isc.sans.edu/podcastdetail/10074>, (Mon, Aug 31st)

Source: SANS Internet Storm Center

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Marginal Coverage Credit Reduces Redundant Exploration in Parallel State-Entropy Optimization

Source: *arXiv cs.LG*

arXiv:2608.27507v1 Announce Type: new Abstract: Policy Gradient for Parallel State Entropy maximization (PGPSE) expands state-space coverage by training independently parameterized policies in replicated copies of the same environment. However, its pooled team-entropy score measures only collective...

Quantization-Triggered Backdoors in Language Models: Cross-Quantizer Transferability and the Validation--Deployment Gap

Source: *arXiv cs.LG*

arXiv:2608.27512v1 Announce Type: new Abstract: Post-training quantization is often treated as a semantically neutral optimization for edge deployment of Large Language Models. When a full-precision source checkpoint is evaluated and quantization is applied downstream without equivalent re-evaluatio...

DAMP: Decay-Aware Mixed-Precision Recurrent-State Quantization

Source: *arXiv cs.LG*

arXiv:2608.27513v1 Announce Type: new Abstract: Softmax attention stores key and value vectors for every preceding token, causing inference memory to grow with sequence length. Recent language models incorporating Gated DeltaNet (GDN) or Kimi Delta Attention (KDA) reduce this cost by replacing the K...

A Deeper Analysis of Block-Sparse Featurizers

Source: *arXiv cs.LG*

arXiv:2608.27515v1 Announce Type: new Abstract: The recently introduced block-sparse featurizer (BSF; Fel et al., 2026) is similar to a sparse autoencoder (SAE), but its atomic unit is a small subspace (a block of directions) rather than a single direction. It is designed for features that live on l...

When Muon Meets Task Interference: A Spectral Perspective on Continual Learning and Model Merging

Source: *arXiv cs.LG*

arXiv:2608.27518v1 Announce Type: new Abstract: Continual learning (CL) and model merging (MM) both aim to obtain a single model that performs well across multiple tasks, challenged respectively by catastrophic forgetting and weight-disentanglement error. In the literature, these difficulties are me...

Dandelion: A Spherical Flower for Neural Simulation of Planetary Dynamics

Source: *arXiv cs.LG*

arXiv:2608.27521v1 Announce Type: new Abstract: Many dynamical processes unfold on the sphere but the default scientific machine learning architectures are Euclidean. Applying these architectures on a regular lat-lon grid causes problems: Cartesian convolutions become distorted at high latitude; 2D...

Self-Explainable Multi-Label Graph Neural Network for Correlated Evidence Attribution

Source: *arXiv cs.LG*

arXiv:2608.27574v1 Announce Type: new Abstract: Multi-label graph learning intends to capture the intrinsic complexity of real-world applications, where one sample is often related to multiple groups or consists of multiple objects. To date, a handful of multi-label graph learning methods exist, but...

Curvature-Aware Radius Shrinkage for Adaptive Nearest Neighbor Classification

Source: *arXiv cs.LG*

arXiv:2608.27634v1 Announce Type: new Abstract: Nearest neighbor classification relies fundamentally on how locality is defined, yet conventional k -NN imposes the same neighborhood cardinality throughout the feature space. This assumption can be inadequate for data whose local geometry varies subs...

More Data Cannot Break a Symmetry: Identifiability by Design

Source: *arXiv cs.LG*

arXiv:2608.27651v1 Announce Type: new Abstract: Unsupervised representational alignment recovers a stimulus-by-stimulus correspondence from geometry alone, but the automorphism group of the stimulus geometry bounds what any such alignment can identify, before data exist. The obvious diagnostic for t...

Unsupervised Continual Learning with Growing Self-Organizing Maps and Synthetic Replay

Source: *arXiv cs.LG*

arXiv:2608.27662v1 Announce Type: new Abstract: This work presents a generative continual learning framework based on growing self-organizing maps (GSOMs) that are augmented with learned distributional statistics as well as encoder-decoder models for class-incremental learning. The proposed approach...

SegBench-GC: Testing Segmentation Invariance in Multi-Step Offline Goal-Conditioned Reinforcement Learning

Source: *arXiv cs.LG*

arXiv:2608.27678v1 Announce Type: new Abstract: Offline goal-conditioned reinforcement learning (GCRL) often uses trajectory structure for future-goal sampling and multi-step targets, yet logged trajectories may be partitioned for administrative reasons that do not correspond to termination. We intr...

SafeStep: An Interactive Demonstration of Semantic Communication for Pedestrian Safety Monitoring

Source: *arXiv cs.LG*

arXiv:2608.27688v1 Announce Type: new Abstract: In this paper, we develop SafeStep, an interactive browser-based semantic communication platform for live pedestrian safety monitoring. SafeStep extracts pedestrian information from four live traffic-camera feeds, transmits it through a semantic commun...

RiskBlend: A Multi-Signal Framework for Test Input Prioritization in Machine Learning Regression Testing

Source: *arXiv cs.LG*

arXiv:2608.27704v1 Announce Type: new Abstract: When machine learning classifiers are retrained, inputs correctly classified by the previous model version may be misclassified by the updated version, creating regression faults that are costly to detect because verifying predictions against ground tr...

DART-FL: Burst-Aware Multitask Federated Learning under Dynamic Inference Demand at the Edge

Source: *arXiv cs.LG*

arXiv:2608.27713v1 Announce Type: new Abstract: Edge intelligence systems increasingly require model training and online inference to coexist on resource-constrained devices, while inference demand can vary substantially across tasks over time. This creates two coupled challenges: sufficient computa...

Beyond Non-IID: Learner--Client Distribution Mismatch in Federated Learning

Source: *arXiv cs.LG*

arXiv:2608.27715v1 Announce Type: new Abstract: Federated learning systems are increasingly deployed to facilitate collaborative model training across a heterogeneous client population. Existing practice mostly implicitly assumes that the aggregated client data distribution is representative of the...

Leveraging a Foundation Model for the EEG-Based Diagnosis of Alzheimer's Disease

Source: *arXiv cs.LG*

arXiv:2608.27719v1 Announce Type: new Abstract: Biological heterogeneity in Alzheimer's Disease (AD) poses a critical diagnostic challenge, particularly for traditional linear methods that fail to capture non-linear neural dynamics. To address this, we propose a diagnostic framework utilizing the La...

Diffusion Distillation for Efficient Weather Ensembles

Source: *arXiv cs.LG*

arXiv:2608.27728v1 Announce Type: new Abstract: Diffusion models generate skillful weather ensembles but require costly iterative sampling. We introduce a supervised energy-distance distillation method that compresses a multi-step diffusion teacher into a single-step student by aligning student fore...

The Calls are Coming from Inside the Model: Investigating Probe-based Detection of Tool-Calling Errors in LLMs

Source: *arXiv cs.LG*

arXiv:2608.27750v1 Announce Type: new Abstract: The hidden states of large language models (LLMs) are known to capture rich information relating to model knowledge and behavior that can be hard to extract from examination of input and output alone. As LLM-based systems increasingly interface with th...

Beyond Search-Imitation: Prior-Directed Exploration for Searchless Chess

Source: *arXiv cs.LG*

arXiv:2608.27757v1 Announce Type: new Abstract: Searchless chess networks reach human master strength from a single forward pass by imitating a stronger teacher: the strongest, Leela Chess Zero's (Lc0) released Chessformer, distills the visit counts of an AlphaZero-style Monte Carlo Tree Search (MCT...

Fast Weight Attention for Continual Learning

Source: *arXiv cs.LG*

arXiv:2608.27763v1 Announce Type: new Abstract: Recurrent fast-weight memories and selective state-space models compress an expanding context into a fixed-size recurrent state, making the state transition an online learning rule. We study this rule under read-after-write autoregressive semantics. Fo...

Initialization Is Critical: Advancing Federated Short-Term Load Forecasting under Load Heterogeneity via Model Initialization

Source: *arXiv cs.LG*

arXiv:2608.27791v1 Announce Type: new Abstract: Short-term load forecasting (STLF) provides essential information for numerous applications in modern power systems. However, accurate STLF often relies on fine-grained smart-meter data from distributed users, raising increasing concerns about data pri...

Node-wise Feature Encoding for Neural Performance Prediction

Source: *arXiv cs.LG*

arXiv:2608.27794v1 Announce Type: new Abstract: As neural networks are increasingly deployed on resource constrained edge devices, accurate prediction of latency and energy is critical for efficient neural architecture search. Existing GNN and transformer based predictors achieve strong results but...

Actionable CBFI: Integrating Structural Decomposition and Causal Counterfactual Recourse for Tabular Machine Learning

Source: *arXiv cs.LG*

arXiv:2608.27821v1 Announce Type: new Abstract: Explainable artificial intelligence (XAI) increasingly calls for actionable counterfactual recourse, yet current methodologies face challenges related to causal invalidity, excessive cognitive burden, and predictive failure. Exhaustive causal search al...

FedEHR-Agents: Federated Agentic Optimization for Automated EHR Modeling

Source: *arXiv cs.LG*

arXiv:2608.27856v1 Announce Type: new Abstract: Recent advances in large language models are enabling autonomous clinical agents to perform increasingly complex electronic health record (EHR) modeling workflows. However, agents deployed at individual hospitals remain constrained by institution-speci...

SOMTab: Set-Order Mamba for Efficient Tabular In-Context Learning

Source: *arXiv cs.LG*

arXiv:2608.27882v1 Announce Type: new Abstract: Tabular foundation models based on in-context learning have recently emerged as strong alternatives to task-specific model fitting. However, the current performance frontier remains dominated by attention-heavy architectures, where attention is used th...

Beyond Pairwise Graphs in Science: Hypergraph Adaptive Wavelet Operators for Parametric PDEs

Source: *arXiv cs.LG*

arXiv:2608.27883v1 Announce Type: new Abstract: Physical systems are often modeled by solution operators that map input fields, parameters, geometries, or past states to steady or future physical states. Learning these maps is difficult, especially for time-dependent systems that must assimilate his...

There and Back Again: Bidirectional Diffusion Bridges for Multimodality Translation

Source: *arXiv cs.LG*

arXiv:2608.27885v1 Announce Type: new Abstract: Multimodality translation (e.g., text-to-image) is a core generative AI task. However, existing approaches (1) follow generative paths that do not directly represent the source modality, limiting the flexibility of some sampling algorithms; and (2) are...

TACIT-Switch: Cost-Aware Model Escalation for LLM Agents from Censored Supervision

Source: *arXiv cs.LG*

arXiv:2608.27911v1 Announce Type: new Abstract: Agents with smaller language-model backbones are less expensive but can drift into persistent failure modes, whereas those with larger backbones are generally more reliable but more costly. This reliability-cost trade-off motivates routing methods that...

TI²PS: A Topology-Informed Inverse Design Framework for Stochastic Multicellular Pattern Formation

Source: *arXiv cs.LG*

arXiv:2608.27931v1 Announce Type: new Abstract: This study proposes a novel framework to estimate parameters for reproducing target multicellular patterns using an agent-based model (ABM). Two major challenges in multicellular ABMs are estimating cell-level parameters (agent-specific variables) and...

Temporal Memory-Aware Online Test-Time Adaptation on Dynamic Graphs

Source: *arXiv cs.LG*

arXiv:2608.27948v1 Announce Type: new Abstract: Test-time adaptation (TTA) on graphs aims to adapt a graph neural network (GNN) that is well-trained on the training graph to the test graph, which involves potential distribution shifts that may harm model generalization and test-time inference. While...

PhyMamba: Physics-Modulated Mamba for Robust Battery Health Prognostics

Source: *arXiv cs.LG*

arXiv:2608.27978v1 Announce Type: new Abstract: Battery health prognostics is a core function in battery management systems (BMSs), yet long-horizon health forecasting from BMS signals remains challenging due to operating-condition dependency and sensor noise. In this paper, we propose PhyMamba, a t...

Is Monte Carlo Tree Search Just Every-Visit Monte Carlo Control?

Source: *arXiv cs.LG*

arXiv:2608.27985v1 Announce Type: new Abstract: Monte Carlo Tree Search (MCTS) and every-visit Monte Carlo (MC) control are usually presented as different methods. MCTS is described in the language of search (selection, expansion, simulation, and backup), whereas MC control is described in the langu...

A Method for Layer Bit-Width Allocation in LLM Quantization via Performance Maximization Under a Quality-Degradation Constraint

Source: *arXiv cs.LG*

arXiv:2608.28003v1 Announce Type: new Abstract: This paper proposes a layer bit allocation method for Gemma-3-1B, formulating the problem as performance maximization (latency decrease) given a degradation budget constraint (allowable level of generation quality loss). This approach is different from...

Exact Risk Ratios for Weighted Data Selection in Linear Regression

Source: *arXiv cs.LG*

arXiv:2608.28007v1 Announce Type: new Abstract: Hanneke, Moran, Shlimovich and Yehudayoff (COLT 2025) posed the following open problem. A selector sees a finite dataset $D \subseteq \mathbb{R}^d \times \mathbb{R}$, picks at most n examples together with nonnegative weights, and hands the weighted...

When Can Conditional Flow Matching Replace Pointwise Negative Log-Likelihood?

Source: *arXiv cs.LG*

arXiv:2608.28010v1 Announce Type: new Abstract: Flow matching enables likelihood-free training, yet alignment methods increasingly reuse conditional flow matching (CFM) losses as endpoint negative log-likelihoods (NLLs) and their old/new differences as log-likelihood ratios. We characterize when the...

Explainable Uncertainty Estimation for Reliable Medical AI

Source: *arXiv cs.LG*

arXiv:2608.28052v1 Announce Type: new Abstract: Artificial intelligence has strong potential to support clinical decision-making, yet its adoption in healthcare remains limited due to a lack of trust. Uncertainty estimation can signal unreliable predictions, and explainable AI (XAI) can clarify how...

Comparing Classical and Quantum Machine Learning for Regression in High Energy Physics Collision Data

Source: *arXiv cs.LG*

arXiv:2608.28084v1 Announce Type: new Abstract: The classification and regression of particle collision events constitute a persistent computational challenge in experimental high energy physics, where large volumes of simulated data must be processed with both speed and precision. This work carries...

Generalized Gibbs Ensemble Weighting for Forecast Combination

Source: *arXiv cs.LG*

arXiv:2608.28116v1 Announce Type: new Abstract: Forecast combination is a reliable way to improve predictive performance when several forecasting models are available. Simple aggregation rules such as the mean, median, trimmed mean, inverse-loss weighting, and exponential weighting are often strong...

VICT: Verifier-Instrumented Credit Tracing for Long-Horizon LLM Agent Reinforcement Learning

Source: *arXiv cs.LG*

arXiv:2608.28128v1 Announce Type: new Abstract: Fine-grained credit assignment is a central challenge in reinforcement learning for long horizon LLM agents. Standard objectives often train from programmatically verifiable terminal rewards by broadcasting each sparse outcome to every action in a traj...

Learning to Difference: Adaptive Reversible Differencing (AdaRDiff) for Time Series Forecasting

Source: *arXiv cs.LG*

arXiv:2608.28134v1 Announce Type: new Abstract: Reliable long-horizon time series forecasting is an important yet difficult problem. Trends and seasonality introduce complex temporal structure that challenges learning-based forecasting models. Differencing, which subtracts nearby past values to remo...

Conditional Diffusion Models for Energy-Efficient Driving

Source: arXiv cs.LG

arXiv:2608.28142v1 Announce Type: new Abstract: Electrification of commercial delivery fleets is shifting fleet routing from distance- and time-based optimization toward energy-aware decision-making. Existing sequence models primarily provide deterministic point estimates or limited uncertainty summ...

The Approximation Rank of Softmax Attention: Sharp Geometric Laws and Robust Interaction Dimension

Source: arXiv cs.LG

arXiv:2608.28150v1 Announce Type: new Abstract: Which geometry controls the rank complexity of normalized softmax attention? We study maximum-row- ℓ_1 approximation rank, exactly the least unrestricted rank preserving every bounded vector-valued output. Two sharp worst-case laws isolate support...

HARTS: Efficient Agentic Reinforcement Learning for Hybrid-Attention Models over Arbitrary Rollout Trees

Source: arXiv cs.LG

arXiv:2608.28158v1 Announce Type: new Abstract: Agentic reinforcement learning (RL) often produces irregular rollout trees with shared histories. Training root-to-leaf trajectories independently recomputes these shared prefixes. Existing systems primarily target full-attention models and lack dense,...

Biologically Inspired Mechanisms for Facilitating Grokking in Multilayer Perceptrons

Source: arXiv cs.LG

arXiv:2608.28184v1 Announce Type: new Abstract: Grokking is a delayed transition from memorization to generalization that is often accompanied by substantial reorganization of internal representations. This paper studies whether biologically inspired mechanisms, many of which are not commonly incorp...

Beyond Flat Netlist: Hierarchical Graph Representation Learning for Scalable Analysis of Sequential Circuits

Source: arXiv cs.LG

arXiv:2608.28188v1 Announce Type: new Abstract: Circuit Representation Learning (CRL) offers a powerful paradigm to guide and optimize core Electronic Design Automation (EDA) tasks, but its practical adoption is hindered by the immense scale of industrial netlists and a failure to explicitly model r...

Performative Privacy: When Differential Privacy Maximizes Utility

Source: *arXiv cs.LG*

arXiv:2608.28198v1 Announce Type: new Abstract: Privacy-preserving learning is often motivated by the idea that protecting users' data can preserve trust and thus participation, improving utility in the long term. However, this claim has not been formalized so far. In parallel, performative learning...

Generalized Context in Cross Attention for Transfer Learning of Disjoint Tabular Data

Source: *arXiv cs.LG*

arXiv:2608.28209v1 Announce Type: new Abstract: Unlike images and text, applying transfer learning to tabular data is challenging due to heterogeneity in feature types, structures, and semantics across disparate domains. Existing methods assume shared features across data tables to enable knowledge...

D-TAIA: Domain-Aware LLM Adaptation for Multi-Task Predictive Process Monitoring

Source: *arXiv cs.LG*

arXiv:2608.28236v1 Announce Type: new Abstract: Predictive Process Monitoring (PPM) enables organizations to forecast future process behavior, such as the next activity and remaining time of ongoing cases. In practice, three conditions cause existing methods to degrade, namely data scarcity, high pr...

Efficient Online Continual Foundation Model Fine-Tuning for Predictive Process Monitoring

Source: *arXiv cs.LG*

arXiv:2608.28237v1 Announce Type: new Abstract: Predictive Process Monitoring (PPM) models are increasingly deployed in dynamic environments where concept drift causes the underlying process distribution to shift over time. While recent work has moved toward online continual learning, existing metho...

Spectral Features Dominate BCG Respiratory-Event Detection: A Large-Scale Patient-Independent Comparison of Feature Groups in Sleep Apnea Patients

Source: *arXiv cs.LG*

arXiv:2608.28242v1 Announce Type: new Abstract: Unobtrusive ballistocardiographic (BCG) sensing is a promising modality for long-term sleep-apnea monitoring, yet it remains unclear which signal features are most discriminative for respiratory-event detection. We present a literature-guided, patient-...

SinkSLOT: Sinkhorn via Sparse Lifted Optimal Transport

Source: *arXiv cs.LG*

arXiv:2608.28262v1 Announce Type: new Abstract: Entropic optimal transport (EOT) has been shown to offer a computationally tractable approximation to exact optimal transport. However, the standard Sinkhorn-Knopp algorithm has two main limitations. First, given discrete measures with N points, each...

Residual-Guided Randomized Neural Networks

Source: *arXiv cs.LG*

arXiv:2608.28267v1 Announce Type: new Abstract: Randomized neural networks enable fast and analytically tractable training by fixing the input to hidden layer parameters at random and learning the output weights in closed form; however, their performance critically depends on a single uninformed dra...

Learning to Transfer Across Modes: Towards Unified Urban Mobility Forecasting

Source: *arXiv cs.LG*

arXiv:2608.28273v1 Announce Type: new Abstract: Urban transportation systems consist of multiple mobility modes that coexist within the same city and exhibit complex interdependencies, leading to correlated demand dynamics across modes. However, forecasting demand jointly across different modes rema...

An algebraic proof of Colombo's difference-power determinant conjecture

Source: *arXiv cs.LG*

arXiv:2608.28274v1 Announce Type: new Abstract: Let $n \geq 2$ be even, let $\lambda = (\lambda_1, \dots, \lambda_n) \in \mathbb{R}^n$ have pairwise distinct coordinates, and define the difference-power matrix $[A_d(\lambda)] := [\text{bigl}[(\lambda_r - \lambda_s)^d \text{bigr}]_{r,s=1}^n, \quad d \in \mathbb{N}.]$ In 192...

Parser States Already Know: Structure-Conditioned KV Persistence for Structured Generation

Source: *arXiv cs.LG*

arXiv:2608.28276v1 Announce Type: new Abstract: Structured generation underpins large language model (LLM) agents that produce JSON, SQL, and function calls, where a single wrong field can cause the downstream action to fail. Constrained decoding already tracks parser transitions to enforce formal v...

VISTA: Verifier-Informed Student-to-Teacher Adaptation for On-Policy Self-Distillation

Source: *arXiv cs.LG*

arXiv:2608.28306v1 Announce Type: new Abstract: On-policy self-distillation (OPSD) improves reasoning by training a problem-only student on its own rollouts using dense token-level supervision from a privileged teacher that also sees a reference solution. However, standard OPSD treats the teacher di...

Deriving Scaling Laws for OpenEuroLLM Models: Learning Rate, Batch Size and Loss

Source: *arXiv cs.LG*

arXiv:2608.28308v1 Announce Type: new Abstract: We study the scaling behavior of learning rate and batch size in pretraining dense large language models on English-prevalent corpora. Beyond scaling \textit{jointly optimal} learning rates and batch sizes, we investigate their \textit{marginal} evolut...

SymboLLM-FE: LLM-Accelerated Symbolic Regression for Automated Feature Engineering on Tabular Data

Source: *arXiv cs.LG*

arXiv:2608.28408v1 Announce Type: new Abstract: Tabular data, as a core data format in machine learning, often lacks the discriminative power needed for high-performance modeling due to insufficient feature informativeness. Automated Feature Engineering (AutoFE) overcomes this by automating feature...

Euclidean Fourier Neural Operators

Source: *arXiv cs.LG*

arXiv:2608.28425v1 Announce Type: new Abstract: Fourier neural operators (FNOs) provide an efficient framework for learning mappings between function spaces as they are, by construction, independent of the grid resolution at which they are trained and evaluated. However, FNOs are not independent of...

Curvature-Conditioned Multiscale Momentum with Sphere Constraints for LLM Pretraining

Source: *arXiv cs.LG*

arXiv:2608.28442v1 Announce Type: new Abstract: Pretraining accounts for a large fraction of the total computational cost in LLM training. However, noise-dominant gradients and the highly ill-conditioned loss landscape bring severe challenges. Although modern adaptive optimizers such as AdamW and Mu...

How Proper Scoring Rules Shape LLM Forecasting

Source: [arXiv cs.LG](#)

arXiv:2608.28482v1 Announce Type: new Abstract: This paper evaluates how reward function choice shapes the performance and behavior of LLM forecasters. We compare five proper scoring rules as training objectives for binary forecasts of resolved real-world events. Although the rules share the same th...

REPLICANT: Learning Policies for Evading and Hardening Malware Detectors

Source: [arXiv cs.LG](#)

arXiv:2608.28499v1 Announce Type: new Abstract: To determine the real-world effectiveness of machine learning based malware detection, it is vital to evaluate its robustness against highly capable adversaries. However, state-of-the-art attacks do not effectively model realistic adversaries, as they...

An Enclosed Mode Is a Gauge Choice: Topology Relative to Reach in Certified Code World Models

Source: [arXiv cs.LG](#)

arXiv:2608.28541v1 Announce Type: new Abstract: A code world model accepted by a sampling gate can be exactly right on everything the gate can see and arbitrarily wrong beyond it. We characterize what a certified model can know, and what its errors can cost, when the omission is an annular freeze mo...

DARTS: Decoder-Aware Representation Tuning via Surgery for Model Merging

Source: [arXiv cs.LG](#)

arXiv:2608.28547v1 Announce Type: new Abstract: Model merging combines multiple task-specific fine-tuned LLMs into a single multi-task model without additional training. However, merged models are known to suffer from representation bias: systematic drift between the merged model's hidden states and...

Advancing Interaction-Sensitive Feature Selection: Novel Relief-Based Algorithms, Expanded Comparisons, and Recommendations for Biomedical Data Mining

Source: [arXiv cs.LG](#)

arXiv:2608.28552v1 Announce Type: new Abstract: As a precursor to high-dimensional biomedical data modeling, reliable feature selection can reduce computational expense, improve modeling performance, and yield simpler, more interpretable models. However, most filter-based feature selection methods s...

Blog: Survey of Optimizers

Source: *arXiv cs.LG*

arXiv:2608.28557v1 Announce Type: new Abstract: Neural-network optimization in 2025-2026 is no longer well described as a succession of new Adam variants. The design space has expanded from coordinates to matrices and layers, from fixed training horizons to policies over time, and from mathematical...

QGPINNs: A Physics-Informed Neural Network Framework for Nonlocal Differential Equations on Quantum Graphs

Source: *arXiv cs.LG*

arXiv:2608.28589v1 Announce Type: new Abstract: We propose QGPINNs, a physics-informed neural network framework developed in PyTorch for the numerical solution of nonlocal differential equations on quantum graphs. The framework is designed as a general computational implementation in which the solut...

Accelerating LLM Inference via Vector Index Based Output Embeddings

Source: *arXiv cs.LG*

arXiv:2608.27460v1 Announce Type: cross Abstract: Large output embedding matrices create a significant memory bandwidth bottleneck during autoregressive decoding, especially for compact LLMs with large multilingual vocabularies. We reformulate the output projection followed by top-k token selection...

SciReC: Diagnostic Evaluation of Multimodal, Multi-Turn Relational Reasoning with Adaptive Interaction

Source: *arXiv cs.LG*

arXiv:2608.27461v1 Announce Type: cross Abstract: Relational reasoning requires the process of perceptual understanding, comparing, and integrating the underlying relationships between concepts. This ability consists of multiple categories, such as analogical, structural, and cause-effect, each capt...

Hypothesize, Evaluate, Refine: A Scientific Agent for PDE Discovery with Unknown Spatial Coefficient Fields

Source: *arXiv cs.LG*

arXiv:2608.27475v1 Announce Type: cross Abstract: Discovering PDEs in heterogeneous media requires jointly identifying the governing operator and the unknown spatial fields that parameterize it. These tasks are coupled: changing field placement changes the differential law, while a sufficiently flex...

Effectiveness of IoT and Deep Learning for Detection and Severity Assessment of *Postelectrotermes militaris* in Tea Plantations

Source: *arXiv cs.LG*

arXiv:2608.27480v1 Announce Type: cross Abstract: Tea plantations are vulnerable to *Postelectrotermes militaris*, commonly known as the Upcountry Live Wood Termite (ULWT), which can cause substantial damage when infestations remain undetected. This study proposes an IoT-enabled acoustic monitoring fr...

Multiscale Community-Based Fingerprinting of Signed Functional Networks

Source: *arXiv cs.LG*

arXiv:2608.27483v1 Announce Type: cross Abstract: Objective: Recent studies demonstrate that functional connectomes contain subject-specific signatures, or $\textit{fingerprints}$, that can identify individuals across repeated sessions and tasks. Existing methods mostly rely on edge-level features tha...

Optimal Transport for Network Comparison: A Review with Machine Learning Applications

Source: *arXiv cs.LG*

arXiv:2608.27500v1 Announce Type: cross Abstract: Network comparison using optimal transport is a growing area of research in network science. Unlike standard graph metrics, optimal transport computes both network dissimilarity and a transport plan that explains how one graph morphs into another. In...

How Do Linear Probes Emerge? A Circuit-Tracing Framework with Concept-Targeted Attribution

Source: *arXiv cs.LG*

arXiv:2608.27510v1 Announce Type: cross Abstract: Transcoder attribution graphs are usually trained to explain why a model assigns high probability to a particular next token. We introduce Concept-Targeted Attribution (CTA), which instead trains attribution graphs with respect to a linear probe dire...

Destroy Me: Automatic Artifact Generation for Histopathology Images

Source: *arXiv cs.LG*

arXiv:2608.27516v1 Announce Type: cross Abstract: Deep learning's diagnostic utility in pathology is constrained by model vulnerability to real-world data imperfections. While current strategies favor "perfect data" by filtering low-quality regions, which can lead to the loss of valuable diagnostic...

Ab initio Modeling of MoS2/Oxide Device Interfaces with Machine Learned Electronic Structures

Source: *arXiv cs.LG*

arXiv:2608.27533v1 Announce Type: cross Abstract: We introduce a new ab initio approach to simulate semiconductor devices that integrates scalable machine-learned (ML) electronic structure models with an advanced quantum transport (QT) solver. The developed framework enables 10,000X speedups over de...

Towards a mathematical theory of superposition

Source: *arXiv cs.LG*

arXiv:2608.27540v1 Announce Type: cross Abstract: We develop a mathematical theory of superposition in neural networks using tools from frame theory and compressed sensing. In our model, a sparse binary vector (x) of active features is encoded through an overcomplete dictionary (W), and feature...

Towards Large-Scale Heterogeneous Data Organization for Scientific Foundation Models: A Nuclear Fusion Case Study

Source: *arXiv cs.LG*

arXiv:2608.27578v1 Announce Type: cross Abstract: Training effective foundation models requires massive and organized datasets, yet scientific domains such as nuclear fusion present unique challenges due to largely heterogeneous and sparse data. Here we characterize the data used in developing such...

Physics-informed learning for the inverse problem in resonant ultrasound spectroscopy

Source: *arXiv cs.LG*

arXiv:2608.27590v1 Announce Type: cross Abstract: Inferring elastic constants from resonant ultrasound spectra is a nonlinear and typically overdetermined inverse problem based on finite spectral data. We formulate the Rayleigh-Ritz inverse problem as a constrained inverse-isospectral problem on the...

Tensor-Accelerated Eager Multi-Resolution Grids for Evolving Large-Scale Substrates

Source: *arXiv cs.LG*

arXiv:2608.27612v1 Announce Type: cross Abstract: In neuroevolution, indirect encoding generates neural network connectivity from a compact genome rather than specifying each connection. ES-HyperNEAT automatically discovers where to place hidden nodes by examining CPPN output patterns: it recursivel...

Quantum SEDONet: Spectrally-Embedded Quantum Deep Operator Networks for Partial Differential Equations

Source: *arXiv cs.LG*

arXiv:2608.27626v1 Announce Type: cross Abstract: Quantum DeepONet accelerates neural-operator inference by evaluating an orthogonally parameterized network on a quantum computer, reproducing in ideal simulation the accuracy of its classical counterpart at asymptotically lower inference cost. Its tr...

Depth-Aware Pothole Detection Using YOLO and RT-DETR at the Edge

Source: *arXiv cs.LG*

arXiv:2608.27633v1 Announce Type: cross Abstract: Pothole detection and its severity measurement is still an important challenges in urban infrastructure management, where late maintenance directly contributes to vehicle damage, road accidents, and escalating repair costs. Existing automated approach...

CARDINAL Predicts Cardiovascular Risk From Non-contrast Cardiac CT

Source: *arXiv cs.LG*

arXiv:2608.27690v1 Announce Type: cross Abstract: Cardiovascular risk prediction remains limited by incomplete clinical data and imaging biomarkers that reduce computed tomography (CT) to a small number of handcrafted features. We developed CARDINAL (Cardiovascular Assessment via Representation lear...

On the Computational and Statistical Efficiency of the Empirical Maximum Entropy on the Mean Method

Source: *arXiv cs.LG*

arXiv:2608.27705v1 Announce Type: cross Abstract: The Maximum Entropy on the Mean (MEM) method provides a flexible computational framework for solving inverse problems by combining data fidelity with entropy-based regularization. In practice, however, the prior distribution is typically unknown but...

Beyond Procrustes distances: a multilinear Gromov-Wasserstein distance capturing chirality

Source: *arXiv cs.LG*

arXiv:2608.27774v1 Announce Type: cross Abstract: Efficiently and robustly analyzing shape data is critical across many scientific disciplines. While chirality is a fundamental property in numerous applications - most notably in molecular science - existing shape analysis metrics fail to distinguish...

Memorization Is Not Extraction: Tight Differential-Privacy Bounds and Audit Blind Spots

Source: *arXiv cs.LG*

arXiv:2608.27782v1 Announce Type: cross Abstract: Memorization in large language models is measured through a zoo of definitions whose formal relations are unknown, and differential privacy (DP) is treated as a proxy against all of them at once. We pin down the exact DP constant for the two that car...

CURA: Certified Runtime Alarms for Computer-Use Agents

Source: *arXiv cs.LG*

arXiv:2608.27808v1 Announce Type: cross Abstract: Self-report is the cheapest oversight channel a deployer has, and on capable computer-use agents (CUAs) it fails precisely where oversight matters. On 361 OSWorld tasks our pipeline, a read-only feasibility gate, a planner, and a GUI executor, reache...

Personalized and Multi-View Representation for Federated Cold-Start Recommendation

Source: *arXiv cs.LG*

arXiv:2608.27826v1 Announce Type: cross Abstract: Federated recommendation (FedRec) enables personalized modeling without centralizing users' interaction histories, but most existing methods assume a fixed item pool and thus overlook the practical cold-item setting where new items continuously arriv...

RealSWE: A Compositional Evaluation of Coding Agents under Realistic User Requests

Source: *arXiv cs.LG*

arXiv:2608.27831v1 Announce Type: cross Abstract: Coding agents are now commonly evaluated on the SWE-bench family of benchmarks, whose tasks are built from curated GitHub issues--long, structured, and information-rich. Real user requests, however, are typically far shorter and less structured. To c...

Anchored Scenario Coverage for Failure-Aware First-Hit Batch Inverse Design

Source: *arXiv cs.LG*

arXiv:2608.27873v1 Announce Type: cross Abstract: Early discovery of at least one valid design satisfying a target requirement is a central objective in failure-prone closed-loop inverse design. A natural batch baseline ranks candidates by a product-form marginal valid-hit score, but selecting the h...

What Do Interaction Representations Actually Measure? Pre-Event Separability in Weakly-Supervised Violence Detection

Source: *arXiv cs.LG*

arXiv:2608.27879v1 Announce Type: cross Abstract: Articulated human pose provides detailed body-configuration information beyond coarse spatial relationships, but whether this detail yields greater discriminative information when the downstream pipeline is held fixed remains unclear. We examine this...

OpenStamp: A Watermark for Open-Source Language Models

Source: *arXiv cs.LG*

arXiv:2608.27899v1 Announce Type: cross Abstract: With the growing prevalence of large language model (LLM) generated content, watermarking is considered a promising approach for attributing text to LLMs and distinguishing it from human-written content. A prominent class of techniques embeds subtle...

Not to Break, but to Attest: Adversarial Probes for Privacy-Preserving LLM Verification

Source: *arXiv cs.LG*

arXiv:2608.27954v1 Announce Type: cross Abstract: Post-deployment changes to large language models can alter behavior while leaving routine outputs largely unchanged, creating a challenge for AI governance when model weights are proprietary. We present a privacy-preserving zk-SNARK-based audit frame...

Twin Worlds: Equivariance-Based Abstention for Evidence-Grounded Reasoning

Source: *arXiv cs.LG*

arXiv:2608.28018v1 Announce Type: cross Abstract: Knowledge-intensive reasoning requires Large Language Models (LLMs) to ground answers in provided evidence. When evidence is insufficient, it is desirable that models abstain rather than confidently generating unsupported answers. Existing abstention...

Characterization of Request and Token Energy Costs for LLM Inference Workloads on GPU Platforms

Source: *arXiv cs.LG*

arXiv:2608.28044v1 Announce Type: cross Abstract: Large language model (LLM) inference serving is priced by tokens, but GPU energy is consumed over inference windows. This accounting mismatch makes token-normalized metrics incomplete, since average output-token energy can decrease even when total re...

Emergent aggregation from collective foraging

Source: *arXiv cs.LG*

arXiv:2608.28046v1 Announce Type: cross Abstract: Collective behaviour in living systems is usually modelled as the outcome of a \emph{direct} social drive: agents are rewarded, or hard-wired, to align with or approach their neighbours. Here we show that aggregation can instead emerge from an \emph{...}

Landau theory of quenched criticality in linear in-context learning

Source: *arXiv cs.LG*

arXiv:2608.28059v1 Announce Type: cross Abstract: In-context learning (ICL) allows a pretrained model to infer a new task from examples supplied in its prompt without updating its parameters. In linear models of ICL, the prediction error develops a double-descent singularity when the number of pretr...

Do Medical Vision Models Reason About Anatomy? Probing the Spatial Inductive Biases of Learned Visual Representations

Source: *arXiv cs.LG*

arXiv:2608.28092v1 Announce Type: cross Abstract: Interpreting a CT scan means comparing structures on either side, judging how far apart organs sit, and knowing where each one belongs. Medical vision encoders are evaluated on diagnostic accuracy, or through assembled multimodal systems where a fail...

CheXtriev: Anatomy-Centered Representation for Case-Based Retrieval of Chest Radiographs

Source: *arXiv cs.LG*

arXiv:2608.28137v1 Announce Type: cross Abstract: We present CheXtriev, a graph-based, anatomy-aware framework for chest radiograph retrieval. Unlike prior methods focussed on global features, our method leverages graph transformers to extract informative features from specific anatomical regions. F...

Under-Mattress Temporal Sensing for Next-Day Agitation Risk Scoring in Dementia Wards

Source: *arXiv cs.LG*

arXiv:2608.28152v1 Announce Type: cross Abstract: Agitation fluctuates over short time horizons in people living with dementia, yet continuous physiological information for anticipating next-day risk is limited. We assessed whether contactless under-mattress signals from the preceding night inform n...

Empowering Local Agriculture: A Deep Learning-Powered Web System for Identifying Bangladeshi Mango Varieties

Source: *arXiv cs.LG*

arXiv:2608.28161v1 Announce Type: cross Abstract: Mango variety identification in Bangladesh is challenging because closely related cultivars can have similar visual characteristics and images are often captured under varying real-world conditions. This work presents a deep learning-based web system...

Conformal Risk-Averse Decision Making with Optimized Certainty Equivalent Risk Control

Source: *arXiv cs.LG*

arXiv:2608.28179v1 Announce Type: cross Abstract: We study risk-averse decision making, in which an agent selects actions while being uncertain about the true system state. The risk is measured via optimized certainty equivalent (OCE) metrics, which generalize popular criteria such as mean-variance...

EXPOSE: Explainable and Domain-Robust Embeddings from Pathology Vision Foundation Models using Sparse Autoencoders

Source: *arXiv cs.LG*

arXiv:2608.28191v1 Announce Type: cross Abstract: Vision Foundation Models (VFMs) are widely used in computational pathology but remain sensitive to domain shifts arising from variations in staining, tissue preparation, and scanner hardware. A key limitation is that VFM embeddings entangle biological...

Explainable Diabetic Retinopathy Classification Using Vision Foundation Models

Source: *arXiv cs.LG*

arXiv:2608.28207v1 Announce Type: cross Abstract: Diabetic retinopathy (DR) is a major cause of preventable blindness, creating a need for accurate and trustworthy automated screening. This study investigates an explainable DR classification framework using vision foundation models and multiple tran...

Stay Within Your Bounds: Distance-Guided Decoding for Guaranteed Context-Free Grammar Compliance

Source: *arXiv cs.LG*

arXiv:2608.28229v1 Announce Type: cross Abstract: Grammar-constrained decoding helps large language models produce syntactically valid structured outputs, such as code, JSON, and SQL. For context-free grammars, many practical decoders enforce local prefix feasibility: each token must keep the curren...

I-FLOP: Fast Learning of Order and Parents from Interventional Data

Source: *arXiv cs.LG*

arXiv:2608.28245v1 Announce Type: cross Abstract: We extend the FLOP (fast learning of order and parents) algorithm recently proposed by Wien\''obst et al. (2026) from observational to interventional data. In particular, we use the interventional BIC score of Hauser and B\"uhlmann (2012), adapting it...

BanglaMed-QA: A Question Answering System for Healthcare Support in Bangla

Source: *arXiv cs.LG*

arXiv:2608.28329v1 Announce Type: cross Abstract: Medical question answering (QA) systems have become crucial tools for providing reliable health information. But they remain very unexplored for low-resource languages like Bangla due to limited datasets and systems tailored to these languages. To ad...

GRACE:Gradient-guided Coreset Selection for LLM Unlearning

Source: *arXiv cs.LG*

arXiv:2608.28361v1 Announce Type: cross Abstract: Machine Unlearning methods for Large Language Models typically assume pre-specified forget and retain sets. In realistic settings, however, requests may provide only a few examples of undesired behavior, requiring forget and retain sets to be inferre...

Real-Time Monitoring of MHD Liquid Metal Flows with Shallow Recurrent Decoders

Source: *arXiv cs.LG*

arXiv:2608.28366v1 Announce Type: cross Abstract: State estimation in magnetohydrodynamic flows is critical for real-time monitoring of liquid metal blankets in tokamak fusion reactors. Due to the multiphysics nature of these phenomena, high-fidelity simulations are computationally prohibitive for r...

Localizing Global Discrepancies: Marginal Contributions and Contextual Anomaly Detection

Source: *arXiv cs.LG*

arXiv:2608.28375v1 Announce Type: cross Abstract: Global goodness-of-fit and discrepancy statistics can establish that a sample departs from a reference distribution without identifying which observations drive the departure. We develop a framework for this localization problem by assigning to each...

Quantum Federated Learning Based on Bures--Uhlmann Geometry for Heterogeneous Noisy Clients

Source: *arXiv cs.LG*

arXiv:2608.28379v1 Announce Type: cross Abstract: Quantum federated learning enables collaborative model training across quantum devices without sharing raw data, and it faces the data and hardware heterogeneity inherent to noisy quantum devices. Utilizing the quantum geometric tensor is a natural r...

Timing-Aware Repurchase Prediction for Web-Scale E-Commerce: Survival Models for Multi-Surface Grocery Recommendation

Source: *arXiv cs.LG*

arXiv:2608.28393v1 Announce Type: cross Abstract: Repurchase recommenders in e-commerce are commonly framed as a binary question asking "will this customer buy this item within W days", a formulation that requires a separately trained model for every horizon of interest. We replace this stack with s...

Post-Training VLMs for Video Mistake Detection

Source: *arXiv cs.LG*

arXiv:2608.28406v1 Announce Type: cross Abstract: Human mistakes are inevitable when following instructions, yet they can lead to severe consequences. As such, there has been an increased interest in developing methods for detecting mistakes in videos, with current methods mostly focusing on closed-...

Sliding-window beats linear attention

Source: *arXiv cs.LG*

arXiv:2608.28444v1 Announce Type: cross Abstract: Due to the nature of quadratic attention, Large Language Models (LLMs) consume a lot of memory and energy. Every new token costs more than the previous one. For each additional token, the keys and values must be stored in memory indefinitely, which i...

Generalized Splines and Gaussian Processes

Source: *arXiv cs.LG*

arXiv:2608.28446v1 Announce Type: cross Abstract: For finite-dimensional linear inverse problems where the variables are Gaussian, it is well-known that the minimum-mean-square error estimator takes the form of a regularized least-squares data fit. In this chapter, we show that this equivalence exte...

Acquire, Repair, Preserve: A Diagnosis-Guided Post-Training Recipe for Small-Model Dialogue Game Agents

Source: *arXiv cs.LG*

arXiv:2608.28458v1 Announce Type: cross Abstract: Interactive dialogue games test a capability that static benchmarks largely leave implicit: a model must carry state across turns, interpret feedback, and choose valid actions under changing constraints. We study this setting in the LM Playschool Cha...

Learning between the peaks: sharp asymptotics for kernel ridge regression under power-law anisotropy

Source: *arXiv cs.LG*

arXiv:2608.28564v1 Announce Type: cross Abstract: We study kernel ridge regression under anisotropic Gaussian data, where the input covariance decays as a power law with exponent $\alpha \geq 0$ for polynomial inner-product kernels. We derive asymptotically sharp expressions for the kernel spectrum a...

On two proofs of d^2 mixing of weighted Dikin walks

Source: *arXiv cs.LG*

arXiv:2608.28566v1 Announce Type: cross Abstract: We study the mixing time of weighted Dikin walks for sampling from exponential distributions on polytopes and truncated positive-semidefinite (PSD) cones. Our first result gives a general total-variation mixing bound under strong self-concordance, \$...

Learning a Size-Weight Frontier for Synthetic-Augmented Inference

Source: *arXiv cs.LG*

arXiv:2608.28576v1 Announce Type: cross Abstract: Synthetic data can improve statistical inference when real data are scarce, but naively treating synthetic samples as real data can introduce bias and lead to unreliable inference. We develop a general framework for synthetic-augmented inference acro...

Aero Hand Open: A Simulation-Ready Tendon-Driven Hand for Dexterous Manipulation Learning

Source: *arXiv cs.LG*

arXiv:2608.28578v1 Announce Type: cross Abstract: Tendon-driven hands are anthropomorphic, and moving the actuators off the joints is what makes a hand of this capability affordable to build. Two effects produce that saving. Routing force through a cable removes the requirement that a motor fit insi...

Trajectory balance: Improved credit assignment in GFlowNets

Source: *arXiv cs.LG*

arXiv:2201.13259v4 Announce Type: replace Abstract: Generative flow networks (GFlowNets) are a method for learning a stochastic policy for generating compositional objects, such as graphs or strings, from a given unnormalized density by sequences of actions, where many possible action sequences may...

Diffusion models as plug-and-play priors

Source: *arXiv cs.LG*

arXiv:2206.09012v4 Announce Type: replace Abstract: We consider the problem of inferring high-dimensional data $\mathbf{x}$ in a model that consists of a prior $p(\mathbf{x})$ and an auxiliary differentiable constraint $c(\mathbf{x}, \mathbf{y})$ on $\mathbf{x}$ given some additional information $\mathbf{y}$

Transformer-Based Autonomous Driving Models and Deployment-Oriented Compression: A Survey

Source: *arXiv cs.LG*

arXiv:2304.10891v4 Announce Type: replace Abstract: Transformer-based models are becoming a central paradigm in autonomous driving because they can capture long-range spatial dependencies, multi-agent interactions, and multimodal context across perception, prediction, and planning. At the same time,...

Let the Flows Tell: Solving Graph Combinatorial Optimization Problems with GFlowNets

Source: *arXiv cs.LG*

arXiv:2305.17010v4 Announce Type: replace Abstract: Combinatorial optimization (CO) problems are often NP-hard and thus out of reach for exact algorithms, making them a tempting domain to apply machine learning methods. The highly structured constraints in these problems can hinder either optimizati...

Joint Bayesian Inference of Graphical Structure and Parameters with a Single Generative Flow Network

Source: *arXiv cs.LG*

arXiv:2305.19366v3 Announce Type: replace Abstract: Generative Flow Networks (GFlowNets), a class of generative models over discrete and structured sample spaces, have been previously applied to the problem of inferring the marginal posterior distribution over the directed acyclic graph (DAG) of a B...

Biases in Expected Goals Models Confound Finishing Ability

Source: *arXiv cs.LG*

arXiv:2401.09940v2 Announce Type: replace Abstract: Expected Goals (xG) has emerged as a popular tool for evaluating finishing skill in soccer analytics. It involves comparing a player's cumulative xG with their actual goal output, where consistent overperformance indicates strong finishing ability....

Improved off-policy training of diffusion samplers

Source: *arXiv cs.LG*

arXiv:2402.05098v5 Announce Type: replace Abstract: We study the problem of training diffusion models to sample from a distribution with a given unnormalized density or energy function. We benchmark several diffusion-structured inference methods, including simulation-based variational approaches and...

Amortizing intractable inference in diffusion models for vision, language, and control

Source: *arXiv cs.LG*

arXiv:2405.20971v3 Announce Type: replace Abstract: Diffusion models have emerged as effective distribution estimators in vision, language, and reinforcement learning, but their use as priors in downstream tasks poses an intractable posterior inference problem. This paper studies amortized sampling...

Meta-Prompt Optimization for LLM-Based Sequential Decision Making

Source: *arXiv cs.LG*

arXiv:2502.00728v2 Announce Type: replace Abstract: Large language models (LLMs) have recently been employed as agents to solve sequential decision-making tasks such as Bayesian optimization and multi-armed bandits (MAB). These works usually adopt an LLM for sequential action selection by providing...

RegCL: Compact Continual SAM Adaptation for Visual Grounding in Multi-Sensorial Media

Source: *arXiv cs.LG*

arXiv:2507.12297v2 Announce Type: replace Abstract: Multi-sensorial media systems, including AR/VR, remote operation, and embodied AI, require visual grounding modules that remain reliable as sensing environments and application domains evolve. The Segment Anything Model (SAM) provides a strong foun...

Attention as Conditioning: What Classical Learning Theory Predicts About Linear Transformers

Source: *arXiv cs.LG*

arXiv:2508.08289v3 Announce Type: replace Abstract: Attention is widely understood as an associative memory, but that description alone does not predict how the memory will behave. Predictive theories do exist, but in the literature on animal learning. We show that the state updates of the major lin...

Class Incremental Continual Learning with Self-Organizing Maps and Synthetic Replay

Source: *arXiv cs.LG*

arXiv:2508.21240v2 Announce Type: replace Abstract: This work introduces a novel generative continual learning framework based on self-organizing maps (SOMs), a brain-inspired natural computing model, extended with learned distributional statistics and encoder-decoder models for class incremental c...

Shift Before You Learn: Enabling Low-Rank Representations in Reinforcement Learning

Source: *arXiv cs.LG*

arXiv:2509.05193v3 Announce Type: replace Abstract: Low-rank structure is a common implicit assumption in many modern reinforcement learning (RL) algorithms. For instance, reward-free and goal-conditioned RL methods often presume that the successor measure admits a low-rank representation. In this w...

Large Reasoning Models Learn Better Alignment from Flawed Thinking

Source: *arXiv cs.LG*

arXiv:2510.00938v3 Announce Type: replace Abstract: Large reasoning models (LRMs) "think" by generating structured chain-of-thought (CoT) before producing a final answer, yet they still lack the ability to reason critically about safety alignment and are easily biased when a flawed premise is inject...

One Model for All: Universal Pre-training for EEG based Emotion Recognition across Heterogeneous Datasets and Paradigms

Source: *arXiv cs.LG*

arXiv:2511.08444v2 Announce Type: replace Abstract: EEG-based emotion recognition is hampered by profound dataset heterogeneity (channel/subject variability), hindering generalizable models. Existing approaches struggle to transfer knowledge effectively. We propose 'One Model for All', a universal p...

Aspiration-based Perturbed Learning Automata in Games with Noisy Utility Measurements. Part A: Stochastic Stability in Non-zero-Sum Games

Source: *arXiv cs.LG*

arXiv:2511.11602v3 Announce Type: replace Abstract: Reinforcement-based learning has attracted considerable attention both in modeling human behavior as well as in engineering, for designing measurement- or payoff-based optimization schemes. Such learning schemes exhibit several advantages, especial...

The Instability of Safety: How Random Seeds and Temperature Expose Inconsistent LLM Refusal Behavior

Source: *arXiv cs.LG*

arXiv:2512.12066v3 Announce Type: replace Abstract: Current safety evaluations of large language models rely on single-shot testing, implicitly assuming that model responses are deterministic and representative of the model's safety alignment. We challenge this assumption by investigating the stabil...

Bayesian Experimental Design for Model Discrepancy Calibration: A Rivalry between Kullback--Leibler Divergence and Wasserstein Distance

Source: *arXiv cs.LG*

arXiv:2601.16425v2 Announce Type: replace Abstract: Designing experiments that systematically gather data from complex physical systems is central to accelerating scientific discovery. While Bayesian experimental design (BED) provides a principled, information-based framework that integrates experim...

Simplex-to-Euclidean Bijection for Conjugate and Calibrated Multiclass Gaussian Process Classification

Source: *arXiv cs.LG*

arXiv:2603.16621v2 Announce Type: replace Abstract: We propose a conjugate and calibrated Gaussian process (GP) model for multi-class classification by exploiting the geometry of the probability simplex. Our approach uses Aitchison geometry to map simplex-valued class probabilities to an unconstrain...

InfoMamba: An Attention-Free Hybrid Mamba-Transformer Model

Source: *arXiv cs.LG*

arXiv:2603.18031v2 Announce Type: replace Abstract: Balancing fine-grained local modeling with long-range dependency capture under computational constraints remains a central challenge in sequence modeling. While Transformers provide strong token mixing, they suffer from quadratic complexity, wherea...

Var-JEPA: A Variational Formulation of the Joint-Embedding Predictive Architecture - Bridging Predictive and Generative Self-Supervised Learning

Source: *arXiv cs.LG*

arXiv:2603.20111v2 Announce Type: replace Abstract: The Joint-Embedding Predictive Architecture (JEPA) is often seen as a non-generative alternative to likelihood-based self-supervised learning, emphasizing prediction in representation space rather than reconstruction in observation space. We argue...

PolicyLong: Towards On-Policy Context Extension

Source: *arXiv cs.LG*

arXiv:2604.07809v2 Announce Type: replace Abstract: Extending LLM context windows is hindered by scarce high-quality long-context data. Recent methods synthesize data with genuine long-range dependencies via information-theoretic verification, selecting contexts that reduce a base model's predictive...

SemEnrich: Self-Supervised Semantic Enrichment of Radiology Reports for Vision-Language Learning

Source: *arXiv cs.LG*

arXiv:2604.09887v2 Announce Type: replace Abstract: Medical vision-language datasets are often limited in size and biased toward negative findings, as clinicians report abnormalities mostly but might omit some positive/neutral findings because they might be considered as irrelevant to the patient's...

Budget-Constrained Causal Bandits: Bridging Uplift Modeling and Sequential Decision-Making

Source: *arXiv cs.LG*

arXiv:2604.26169v2 Announce Type: replace Abstract: Treatment allocation under budget constraints is a central challenge in digital advertising. The standard approach trains an offline uplift model on historical data, then solves a constrained optimization to allocate budget. This fails in cold-start...

ABC: Any-Subset Autoregression via Non-Markovian Diffusion Bridges in Continuous Time and Space

Source: *arXiv cs.LG*

arXiv:2604.27443v3 Announce Type: replace Abstract: Generating continuous-time, continuous-space stochastic processes (e.g., videos, weather forecasts) conditioned on partial observations (e.g., first and last frames) is a fundamental challenge. Existing approaches, (e.g., diffusion models), suffer...

ImplicitTerrainV2: Wavelet-Guided Spatially Adaptive Neural Terrain Representation

Source: *arXiv cs.LG*

arXiv:2605.22556v3 Announce Type: replace Abstract: Digital elevation models (DEMs) underpin terrain analysis in Geographic Information Systems (GIS), but commonly as raster representation, they rely on interpolation for off-grid sampling and finite-difference operators for derivative-based analysis...

Learned Relay Representations for Forward-Thinking Discrete Diffusion Models

Source: *arXiv cs.LG*

arXiv:2605.22967v3 Announce Type: replace Abstract: When Masked Diffusion Models (MDMs) generate sequences through iterative refinement, the rich internal computation over masked positions is discarded, forcing every subsequent refinement step to recompute the valuable internal information stored as...

More Expressive Feedforward Layers: Part I. Token-Adaptive Mixing of Activations

Source: *arXiv cs.LG*

arXiv:2605.26647v2 Announce Type: replace Abstract: Feedforward network (FFN) layers account for a large fraction of parameters and nonlinear expressivity in Transformer-based large language models (LLMs). Despite the evolution from ReLU and GELU to gated variants such as SwiGLU, most FFN designs st...

Negligible in Size, Significant in Effect: On Scale Vectors in Large Language Models

Source: *arXiv cs.LG*

arXiv:2605.26895v2 Announce Type: replace Abstract: Normalization layers in modern large language models (LLMs) consist of a deterministic normalization operation and a learnable scale vector. While the normalization operation has been extensively studied, the scale vector remains poorly understood...

LongDS-Bench: On the Failure of Long-Horizon Agentic Data Analysis

Source: *arXiv cs.LG*

arXiv:2605.30434v2 Announce Type: replace Abstract: Real-world data analysis is inherently iterative, yet existing benchmarks mostly evaluate isolated or short interactive tasks, leaving agents' ability to track evolving analytical context over long horizons untested. We introduce LongDS, a benchmar...

Can Subgraph Explanations Be Weaponized to Steal Graph Neural Networks?

Source: *arXiv cs.LG*

arXiv:2605.30470v2 Announce Type: replace Abstract: Graph Machine Learning as a Service (GMLaaS) platforms increasingly implement explainability interfaces to meet regulatory transparency requirements. However, this transparency creates exploitable vulnerabilities for model extraction attacks. We pr...

ERP-XTTN: Interpretable Prototype-Guided Cross-Attention for Cross-Subject ERP Classification

Source: *arXiv cs.LG*

arXiv:2606.02939v2 Announce Type: replace Abstract: Interpretable brain-computer interface classifiers that generalize across subjects without calibration remain an open challenge. We evaluated whether prototype-based cross-attention can provide competitive, inherently interpretable ERP classificati...

The Discrete-Log Clock: How a Transformer Learns Modular Multiplication

Source: *arXiv cs.LG*

arXiv:2606.17399v2 Announce Type: replace Abstract: When small transformers grok modular multiplication, prior work reports that the learned embedding has a "dense" Fourier spectrum requiring all frequencies. This contrasts with modular addition, where only a sparse set of key frequencies suffices....

SpecGradFilter: A Spectral Gradient Filtering Framework for Taming Federated Heterogeneity

Source: *arXiv cs.LG*

arXiv:2607.04189v2 Announce Type: replace Abstract: Federated Learning (FL) is fundamentally challenged by statistical heterogeneity, where non-identically distributed (non-IID) data induces client drift that severely hampers global convergence. While existing approaches attempt to mitigate this dri...

On the Depth Scalability of Logic Gate Networks

Source: *arXiv cs.LG*

arXiv:2607.21633v3 Announce Type: replace Abstract: Logic Gate Networks (LGNs) compute through compositions of Boolean operations, yet existing LGNs do not reliably benefit from increased depth. We identify two causes: optimization collapse and topology-induced degradation of output-specific credit...

Locked Evaluation Surfaces: Transfer Failure and Sampling-Depth Entanglement in CRISPRi Perturbation-Effect Prediction

Source: *arXiv cs.LG*

arXiv:2608.00152v2 Announce Type: replace Abstract: Predicting how held-out target genes respond to CRISPRi perturbation, and whether such predictions transfer across biological screens, is hard to evaluate: a representation can be informative within one screen yet fail across screens, while endpoint...

ED-CSP: Crystal Structure Prediction from Electron Diffraction

Source: *arXiv cs.LG*

arXiv:2608.06448v3 Announce Type: replace Abstract: Recovering a periodic 3D crystal structure from sparse, unindexed electron diffraction (ED) observations is a challenging generative inverse problem. Existing ED-based learning methods mainly predict crystallographic labels, reconstruct structures...

Recirculation

Source: *arXiv cs.LG*

arXiv:2608.17981v2 Announce Type: replace Abstract: We describe an inference-time architectural enhancement for off-the-shelf foundation models that markedly reduces perplexity and boosts accuracy across generation and reasoning tasks. Our approach incurs essentially no additional latency during gen...

How Far Should Tokenization Go? Predictive Effectiveness and Relational Losslessness

Source: *arXiv cs.LG*

arXiv:2608.18025v2 Announce Type: replace Abstract: GPT-style models have achieved remarkable success with finite vocabularies of reusable tokens, making the token interface a central component of modern sequence modeling. Symbolic music appears naturally compatible with this paradigm: it consists o...

How Architecture and Training Affect TPC Representations Across Experiments

Source: *arXiv cs.LG*

arXiv:2608.21756v2 Announce Type: replace Abstract: Deep-learning efforts have increasingly shifted toward foundation model approaches. In experimental physics, this allows models and learned representations to be reused beyond the experiments in which they were developed. This work evaluates the re...

RIBOSPAN: A Long-Context RNA Foundation Model for Versatile RNA Modeling

Source: *arXiv cs.LG*

arXiv:2608.22849v2 Announce Type: replace Abstract: Full-length RNAs, particularly messenger RNAs, often exceed the context lengths used to pretrain existing RNA foundation models, limiting complete-transcript modeling at single-nucleotide resolution. We present RIBOSPAN, a 1.61-billion-parameter bi...

JEPA-x: Cross-Predictive Physics Grounding for Forecastable Latent Dynamics

Source: *arXiv cs.LG*

arXiv:2608.24044v3 Announce Type: replace Abstract: Latent world models plan by predicting how candidate actions advance learned latent dynamics. In self-predictive models, however, the encoder and predictor are optimized jointly and can co-adapt to latent transitions that are easy to predict but we...

On-policy Distillation with Verifiable Reward

Source: *arXiv cs.LG*

arXiv:2608.24696v2 Announce Type: replace Abstract: Reinforcement Learning with Verifiable Rewards (RLVR) and on-policy distillation (OPD) have become two widely adopted paradigms for post-training large language models. However, RLVR suffers from sparse task-level feedback, while OPD provides dense...

Trust the Mass: Forced Weights in KV-Cache Eviction

Source: *arXiv cs.LG*

arXiv:2608.25230v2 Announce Type: replace Abstract: Every deployed sparse-attention or KV-cache-eviction rule keeps a subset of the keys, discards the rest, and renormalizes the attention weights over the kept set. Enumerating the exact best subset under that constraint on 168×192 attention rows...

TraceML: An Empirical Analysis of Human-Agent Planning in Machine Learning Development

Source: *arXiv cs.LG*

arXiv:2608.26086v2 Announce Type: replace Abstract: Large language models write correct code for isolated problems but remain far weaker at autonomous machine-learning development, where an agent must revise data pipelines, models, and validation over hours of feedback, and on most competitions stil...

Neural Regression with Embeddings for Numerical Attribute Prediction in Knowledge Graphs

Source: *arXiv cs.LG*

arXiv:2608.26729v2 Announce Type: replace Abstract: In recent years, transductive knowledge graph embedding models have been applied to tasks such as link prediction and query answering. Although knowledge graphs often contain rich numerical attributes, most embedding models neglect them, limiting t...

Accurate prediction is not profitable advice: profit-based evaluation of machine learning nitrogen recommendations in winter wheat

Source: *arXiv cs.LG*

arXiv:2608.27205v2 Announce Type: replace Abstract: Nitrogen rates for winter wheat are set before the season, under unknown prices and weather. The standard UK advice does not respond to prices, yet recent price swings moved the most profitable rate by tens of kilograms per hectare. Machine learnin...

Understanding Evolution Strategies for LLM Reasoning: Broader Reasoning Coverage than GRPO

Source: *arXiv cs.LG*

arXiv:2608.27351v2 Announce Type: replace Abstract: Evolution Strategies (ES) have recently emerged as a memory-efficient post-training paradigm for LLM reasoning. However, the optimization behavior of ES remains understudied, making it hard to define its advantage scope compared to mainstream post-...

Rethinking Speaker Embeddings for Speech Generation: Sub-Center Modeling for Capturing Intra-Speaker Diversity

Source: *arXiv cs.LG*

arXiv:2407.04291v4 Announce Type: replace-cross Abstract: Modeling speech variation is key to natural, expressive generation. Speaker embeddings are commonly used to condition personalized speech systems, but they are typically trained for speaker recognition, where intra-speaker variability is supp...

Mixture of Multicenter Experts in Multimodal AI for Debiased Radiotherapy Target Delineation

Source: *arXiv cs.LG*

arXiv:2410.00046v4 Announce Type: replace-cross Abstract: Clinical decision-making reflects diverse strategies shaped by regional patient populations and institutional protocols. However, most existing medical artificial intelligence (AI) models are trained on highly prevalent data patterns, which r...

Off the Normal Path: Learning Spatial Density Models of Node Mobility

Source: *arXiv cs.LG*

arXiv:2411.10997v2 Announce Type: replace-cross Abstract: We consider the problem of learning models of spatial density functions, representing the steady-state density of mobile nodes moving on a two-dimensional terrain. Deriving such models can assist in network design and optimization problems, e...

Ampere: Communication-Efficient and High-Accuracy Split Federated Learning

Source: *arXiv cs.LG*

arXiv:2507.07130v2 Announce Type: replace-cross Abstract: A Federated Learning (FL) system collaboratively trains neural networks across devices and a server but is limited by significant on-device computation costs. Split Federated Learning (SFL) systems mitigate this by offloading a block of layer...

Automatic Pronunciation Error Detection and Correction of the Holy Quran's Learners Using Deep Learning

Source: *arXiv cs.LG*

arXiv:2509.00094v2 Announce Type: replace-cross Abstract: Assessing spoken language is challenging, and quantifying pronunciation metrics for machine learning models is even harder. However, for the Holy Quran, this task is enabled by the rigorous recitation rules (Tajweed) established through the e...

Probabilistic Symbolic Regression for Equation Discovery via Operator-induced and Regularized Symbolic Forests

Source: *arXiv cs.LG*

arXiv:2509.19710v3 Announce Type: replace-cross Abstract: Symbolic regression has emerged as a powerful tool for artificial intelligence-driven scientific discovery by learning interpretable analytical expressions that reveal governing relationships directly from data. Existing methods, however, oft...

Examining the robustness of Physics-Informed Neural Networks to noise for Inverse Problems

Source: *arXiv cs.LG*

arXiv:2509.20191v2 Announce Type: replace-cross Abstract: Approximating solutions to partial differential equations (PDEs) is fundamental for the modeling of dynamical systems in science and engineering. Physics-informed neural networks (PINNs) are a recent machine learning-based approach, for which...

OceanGym: A Benchmark Environment for Underwater Embodied Agents

Source: *arXiv cs.LG*

arXiv:2509.26536v3 Announce Type: replace-cross Abstract: We introduce OceanGym, the first comprehensive benchmark for ocean underwater embodied agents, designed to advance AI in one of the most demanding real-world environments. Unlike terrestrial or aerial domains, underwater settings present extr...

GREAT: Generalizable Backdoor Attacks in RLHF via Emotion-Aware Trigger Synthesis

Source: *arXiv cs.LG*

arXiv:2510.09260v3 Announce Type: replace-cross Abstract: Recent work has shown that RLHF is highly susceptible to backdoor attacks. However, existing methods often rely on rare tokens or fixed triggers, limiting their impact in realistic scenarios. In this work, we develop GREAT, a novel framework...

Multilingual Lexical Feature Analysis of Spoken Language for Predicting Major Depression Symptom Severity

Source: *arXiv cs.LG*

arXiv:2511.07011v2 Announce Type: replace-cross Abstract: Background: Remotely captured spoken language could provide objective, regular indicators of depression symptom severity. However, research to date has largely used non-clinical, cross-sectional written language and complex machine learning (...)

Think-at-Hard: Dynamic Looped Transformers for Improved Reasoning

Source: *arXiv cs.LG*

arXiv:2511.08577v4 Announce Type: replace-cross Abstract: Improving the reasoning abilities of Large Language Models (LLMs), especially under parameter constraints, is crucial for real-world applications. Looped transformers address this by performing multiple latent iterations to refine each token...

Prequential posteriors

Source: *arXiv cs.LG*

arXiv:2511.17721v2 Announce Type: replace-cross Abstract: Data assimilation is a fundamental task in updating forecasting models upon observing new data, with applications ranging from weather prediction to online reinforcement learning. Deep generative forecasting models (DGFM) have shown excellen...

Aligning Agentic World Models via Knowledgeable Experience Learning

Source: *arXiv cs.LG*

arXiv:2601.13247v2 Announce Type: replace-cross Abstract: Current Large Language Models (LLMs) exhibit a critical modal disconnect: they possess vast semantic knowledge but lack the procedural grounding to respect the immutable laws of the physical world. Consequently, while these agents implicitly...

Learning Fast Monomial Orders for Gröbner Basis Computations

Source: *arXiv cs.LG*

arXiv:2602.02972v2 Announce Type: replace-cross Abstract: The efficiency of Gröbner basis computation, the standard engine for solving systems of polynomial equations, depends on the choice of monomial ordering. Despite a near-continuum of possible monomial orders, most implementations rely on sta...

SCALE: Self-uncertainty Conditioned Adaptive Looking and Execution for Vision-Language-Action Models

Source: *arXiv cs.LG*

arXiv:2602.04208v3 Announce Type: replace-cross Abstract: Vision-Language-Action (VLA) models have emerged as a promising paradigm for general-purpose robotic control, with test-time scaling (TTS) gaining attention to enhance robustness beyond training. However, existing TTS methods for VLAs require...

Robust Assortment Optimization from Observational Data

Source: *arXiv cs.LG*

arXiv:2602.10696v3 Announce Type: replace-cross Abstract: Assortment optimization is a fundamental challenge in modern retail and recommendation systems, where the goal is to select a subset of products that maximizes expected revenue under complex customer choice behaviors. While recent advances in...

Mine and Refine: Optimizing Graded Relevance in E-commerce Semantic Search Retrieval

Source: *arXiv cs.LG*

arXiv:2602.17654v2 Announce Type: replace-cross Abstract: Embedding-based retrieval (EBR) for large-scale e-commerce search faces three intertwined challenges: graded (non-binary) relevance where engagement signals are noisy and intent-varying while business relevance guidelines admit acceptable-but...

FlowCorrect: Efficient Interactive Correction of Generative Flow Policies for Robotic Manipulation

Source: [arXiv cs.LG](#)

arXiv:2602.22056v3 Announce Type: replace-cross Abstract: Generative manipulation policies can fail catastrophically under deployment-time distribution shift, yet many failures are near-misses: the robot reaches almost-correct poses and would succeed with a small corrective motion. We propose FlowCo...

Agentic-Kube: A Graph-Enhanced Multi-Agent Reinforcement Learning Framework for Multi-Objective Kubernetes Scheduling

Source: [arXiv cs.LG](#)

arXiv:2603.12031v4 Announce Type: replace-cross Abstract: Cloud-native container orchestration requires resource schedulers capable of balancing infrastructure expenditure, fault resilience, and node utilisation. Conventional reinforcement learning approaches typically rely on monolithic single-agen...

The Autonomy Tax: Defense Training Breaks LLM Agents

Source: [arXiv cs.LG](#)

arXiv:2603.19423v3 Announce Type: replace-cross Abstract: Large language model (LLM) agents increasingly rely on external tools (file operations, API calls, database transactions) to autonomously complete complex multi-step tasks. Practitioners deploy defense-trained models to protect against prompt...

Camera-Agnostic Pruning of 3D Gaussian Splats via Descriptor-Based Beta Evidence

Source: [arXiv cs.LG](#)

arXiv:2603.21933v3 Announce Type: replace-cross Abstract: The pruning of 3D Gaussian splats is essential for reducing their complexity to enable efficient storage, transmission, and downstream processing. However, most of the existing pruning strategies depend on camera parameters, rendered images,...

Deflation-PINNs: Learning Multiple Solutions for PDEs and Landau-de Gennes

Source: [arXiv cs.LG](#)

arXiv:2603.27936v3 Announce Type: replace-cross Abstract: Nonlinear Partial Differential Equations (PDEs) are ubiquitous in mathematical physics and engineering. Although Physics-Informed Neural Networks (PINNs) have emerged as a powerful tool for solving PDE problems, they typically struggle to ide...

Prompts Without Evidence: How Neuroimaging Mentions Shift Clinical Vision-Language Model Predictions

Source: *arXiv cs.LG*

arXiv:2603.28387v3 Announce Type: replace-cross Abstract: Trustworthy clinical AI must use real evidence and avoid relying on surface-level artifacts. We evaluate 12 open-weight vision-language models (VLMs) on two clinical neuroimaging cohorts for binary classification of affective disorders and co...

Beyond Output Correctness: Benchmarking and Evaluating Large Language Model Reasoning in Coding Tasks

Source: *arXiv cs.LG*

arXiv:2604.12379v2 Announce Type: replace-cross Abstract: Large language models (LLMs) increasingly rely on explicit reasoning to solve coding tasks, yet evaluating the quality of this reasoning remains challenging. Existing reasoning evaluators are not designed for coding, and current benchmarks fo...

G-Loss: Graph-Guided Fine-Tuning of Language Models

Source: *arXiv cs.LG*

arXiv:2604.25853v4 Announce Type: replace-cross Abstract: Traditional loss functions, including cross-entropy, contrastive, triplet, and supervised contrastive losses, used for fine-tuning pre-trained language models such as BERT, operate only within local neighborhoods and fail to account for the...

DiffAnon: Diffusion-based Prosody Control for Voice Anonymization

Source: *arXiv cs.LG*

arXiv:2604.26281v2 Announce Type: replace-cross Abstract: To preserve or not to preserve prosody is a central question in voice anonymization. Prosody conveys meaning and affect, yet is tightly coupled with speaker identity. Existing methods either discard prosody for privacy or lack a principled me...

D3-Gym: Constructing Real-World Verifiable Environments for Data-Driven Discovery

Source: *arXiv cs.LG*

arXiv:2604.27977v3 Announce Type: replace-cross Abstract: Despite recent progress in language models and agents for scientific data-driven discovery, advancing their capabilities is held back by the absence of verifiable environments representing real-world scientific tasks. To fill this gap, we int...

SkillSafetyBench: Evaluating Agent Safety under Skill-Facing Attack Surfaces

Source: arXiv cs.LG

arXiv:2605.12015v3 Announce Type: replace-cross Abstract: Reusable skills are becoming a common interface for extending large language model agents, packaging procedural guidance with access to files, tools, memory, and execution environments. However, this modularity introduces attack surfaces that...

Online Learning-to-Defer with Varying Experts

Source: arXiv cs.LG

arXiv:2605.12340v5 Announce Type: replace-cross Abstract: Learning-to-Defer (L2D) methods route each query either to a predictive model or to external experts. Real-world deployments require handling streaming data, changing expert availability, shifting expert reliability, and feedback observed onl...

WINO: A Weak-Form Physics Informed Neural Operator for Hyperelasticity on Variable Domains

Source: arXiv cs.LG

arXiv:2605.24651v3 Announce Type: replace-cross Abstract: We propose a Weak-form Physics-Informed Neural Operator (WINO), a data-free framework that combines the efficiency of neural operators with the geometric flexibility of the φ -finite element method (φ -FEM). φ -FEM is an...

ToolSense: A Diagnostic Framework for Auditing Parametric Tool Knowledge in LLMs

Source: arXiv cs.LG

arXiv:2606.12451v2 Announce Type: replace-cross Abstract: Large language models deployed as agents over large tool catalogs face a critical tool-retrieval bottleneck. As embedding-based retrieval approaches rely on compact encoders that may under-capture specialized tool semantics, parametric tool r...

RecourseBench: A Modular Framework for Reproducible Algorithmic Recourse Evaluation

Source: arXiv cs.LG

arXiv:2606.16113v2 Announce Type: replace-cross Abstract: Algorithmic recourse methods provide counterfactual explanations that inform individuals of the actions required to overturn an unfavorable model decision. Despite rapid methodological progress, principled comparison remains elusive; existing...

TokenPilot: Cache-Efficient Context Management for LLM Agents

Source: *arXiv cs.LG*

arXiv:2606.17016v2 Announce Type: replace-cross Abstract: As LLM agents are deployed in long-horizon sessions, context accumulation drives up inference costs. Existing approaches utilize text pruning or dynamic memory eviction to minimize token footprints; however, their unconstrained sequence mutat...

Closing the Operational Gap in Semantic Caching

Source: *arXiv cs.LG*

arXiv:2606.19719v3 Announce Type: replace-cross Abstract: Semantic caching cuts LLM inference costs by serving a cached response to semantically similar queries. Standard practice evaluates these systems using PR-AUC, a metric that only measures how well scores rank and ignores whether they are usab...

An End-to-End Hybrid Quantum--Classical Sampling Workflow for Discrete Markov Random Fields: A Reproducible Case Study

Source: *arXiv cs.LG*

arXiv:2607.09893v2 Announce Type: replace-cross Abstract: Sampling from discrete Markov random fields (MRFs) is a hard problem. We study amplitude-encoded i.i.d. sampling for small MRFs where 2^n target probabilities are precomputed classically. This removes quantum exponential speedup but allows...

Robust Chance-Constrained Optimization using a Continuous Parameter Space Wasserstein-2 Ambiguity Set of Gaussian Mixtures

Source: *arXiv cs.LG*

arXiv:2607.17018v2 Announce Type: replace-cross Abstract: We study distributionally robust linear chance-constrained problems in which uncertainty is modeled by a Gaussian mixture model (GMM). Finite-support distributionally robust (FDR) formulations, widely used in data-driven robust optimization,...

Where Steering Signals Come From: Activation Source Selection in Activation Steering

Source: *arXiv cs.LG*

arXiv:2607.25270v2 Announce Type: replace-cross Abstract: Activation steering controls language models by adding vectors or features to hidden states at inference time, but the upstream source of these steering signals is often treated as a secondary detail. We study this source choice as activation...

Establishing Boundary KKT Convergence of Mirror Descent through Reparameterization

Source: *arXiv cs.LG*

arXiv:2608.07248v3 Announce Type: replace-cross Abstract: Sequence convergence to a boundary Karush--Kuhn--Tucker (KKT) point has long remained unclear for nonconvex mirror descent with Legendre kernels. The difficulty arises from the blow-up of the gradient of the Legendre kernel at the boundary. R...

JuryProbe: An Empirical Consensus-Risk Diagnostic for Routing Reference-Free Factuality Judge Panels to Grounded Verification

Source: *arXiv cs.LG*

arXiv:2608.20607v2 Announce Type: replace-cross Abstract: Panels of inexpensive LLM judges increasingly make accept-or-escalate decisions. In factuality settings, accepting a claim because several reference-free judges agree can create a hidden risk: agreement may reflect shared false-negative blind...

What Neural Network Field Theory Can and Cannot Realise on a Computer

Source: *arXiv cs.LG*

arXiv:2608.21523v2 Announce Type: replace-cross Abstract: One aim of neural network field theory is to put a quantum or effective field theory on a computer, with the network ensemble itself as the theory. We ask how far that aim can be pushed for a function class regular enough to be computed with....

GAN-Diff : Coupling Pretrained WGAN-GP Features with Conditional Diffusion U-Nets

Source: *arXiv cs.LG*

arXiv:2608.22272v2 Announce Type: replace-cross Abstract: Generative adversarial networks (GANs) can provide efficient image generation, while diffusion models offer high-quality image restoration but require iterative sampling. This paper presents a hybrid GAN-guided diffusion framework that uses a...

Semantic Overlays: Mitigating Prompt Injection with Annotations Beyond Tokens and Steering Vectors

Source: *arXiv cs.LG*

arXiv:2608.23873v2 Announce Type: replace-cross Abstract: Everything a language model sees is tokens. The serving stack knows what each span is -- user input, tool output, instructions -- but the model must keep track of that itself, and can lose track or be confused: text can be written to read lik...

Understanding ChatGPT Work

Source: Simon Willison's Blog

OpenAI announced ChatGPT Work on July 9th, and have been furiously iterating on it ever since. It is an extraordinarily confusing and very powerful product. Here's what I've figured out about it so far. ChatGPT Work is actually two products The more interesting version of ChatGPT Work is the one tha...